THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The audible range of frequencies for humans is about: 1

(a) 2 Hz to 200 Hz **(b)** 200 Hz to 2000 Hz
(c) 20 kHz to 200 kHz **(d)** 20 Hz to 20000 Hz

2. SONAR is used to: 1

(a) send radio signals **(b)** measure heat
(c) find the depth of the sea **(d)** see in the dark

3. A region in a sound wave where the particles are far apart is a: 1

(a) rarefaction **(b)** crest **(c)** node **(d)** compression

4. Sounds of frequencies below 20 Hz are called: 1

(a) ultrasonic **(b)** audible **(c)** supersonic **(d)** infrasonic

5. Ultrasound is used in medicine for: 1

(a) radio communication **(b)** cutting with light
(c) imaging internal organs **(d)** heating bones only

6. The distance between two consecutive compressions is the: 1

(a) time period **(b)** wavelength **(c)** amplitude **(d)** frequency

7. Sounds of frequencies above 20000 Hz are called: 1

(a) infrasonic **(b)** subsonic **(c)** audible **(d)** ultrasonic

8. Multiple reflections of sound in a big hall cause: 1

(a) interference only **(b)** reverberation
(c) refraction **(d)** dispersion

9. The number of oscillations per second is the: 1

(a) speed **(b)** wavelength **(c)** amplitude **(d)** frequency

10. Sound travels fastest in: 1

(a) solids **(b)** vacuum **(c)** liquids **(d)** gases

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is an echo? State the condition to hear it. 2

12. State two uses of ultrasound. 2

13. Define frequency, wavelength and amplitude of a sound wave. 2

14. What is echolocation? Name two animals that use it. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an activity to show that sound is produced by vibrations using a tuning fork. 3

16. Why is the speed of sound greater in solids than in gases? Give the speeds in air, water and steel (approx). 3

17. Differentiate between a transverse wave and a longitudinal wave with one example each. 3

OR

Explain how sound propagates in air using compressions and rarefactions. 3

18. Why do we hear the sound of thunder after seeing lightning? Calculate the distance of lightning if the thunder is heard 5 s later (v = 340 m/s). 3

OR

Draw in words a displacement-distance graph of a sound wave and label the crest, trough, amplitude and wavelength. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Describe the characteristics of sound: pitch, loudness, quality, and relate them to the wave properties, with the effect on the shape of the wave. 5

OR

Compare the speed of sound in solids, liquids and gases with examples and explain the effect of temperature and the medium. 5

20. Explain the echo and its conditions. A man stands 85 m from a wall and claps. Find the time for the echo (v = 340 m/s) and say if it can be heard distinctly. What is the minimum distance for a distinct echo? 5

OR

Describe an activity to show that sound needs a medium (bell jar experiment) and write the observation and conclusion. 5

END OF QUESTION PAPER — Total: 40 Marks

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